Synoptic CB: Manual Redox Probes

2025 Samples

2026-04-01

Measurements taken with SWAP Redox Probes & Calomel Ref probe

Protocol: https://docs.google.com/document/d/

1dL2ZPhLFBAZJmVBCfKxup_KOYY96qA5IWfJmQen50pY/edit

Units = mV

##Add Required Packages

##Setup

```
###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
dat <- read.csv("Raw Data/COMPASS_Synoptic_CB_Redox_2025_Raw.csv")
soildat <- read.csv("Raw Data/CB_Soil_pH.csv")

#file path and name of processed data file
processed_file_name = "Processed Data/COMPASS_Synoptic_CB_Redox_2025_processed.csv"

##Fix Sample Zone and Site naming
soildat <- soildat %>%
  mutate(Zone = case_when(Site == "SWH" & Zone == "Upland" ~ "Swamp",
                          Site == "SWH" & Zone == "Upland-Control" ~ "Upland",
                          TRUE ~ Zone),
         Site = ifelse(Site == "Gcrew", "GCW", Site))

##Std error function
std.error <- function (x, na.rm=FALSE){
  sqrt(var(x, na.rm = na.rm) / sum(!is.na(x)))}
```

Adjust Raw Readings by pH

Need to adjust for pH:

redox = redoxi + [(ph - 5)59 + 224] This is the equation from Megonigal,

Patrick, & Faulkner 1993

They used a calomel electrode and used 224 to adjust to H2,

we use a KCl ref probe, so we need to add 202 instead

##Write Out Adjusted & Avg'd Data

Vizualize Data